

Using the Expolinear Growth Equation for Modelling Crop Growth in Year-round Cut Chrysanthemum

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The aim of this study was to predict crop growth of year-round cut chrysanthemum (*Chrysanthemum morifolium* Ramat.) based on an empirical model of potential crop growth rate as a function of daily incident photosynthetically active radiation (PAR, MJ m⁻² d⁻¹), using generalized estimated parameters of the expolinear growth equation. For development of the model, chrysanthemum crops were grown in four experiments at different plant densities (32, 48, 64 and 80 plants m⁻²), during different seasons (planting in January, May–June and September) and under different light regimes [natural light, shading to 66 and 43 % of natural light, and supplementary assimilation light (ASS, 40–48 µmol m⁻² s⁻¹)]. The expolinear growth equation as a function of time (EXPOT) or as a function of incident PAR integral (EXPOPAP) effectively described periodically measured total dry mass of shoot ($R^2 > 0.98$). However, growth parameter estimates for the fitted EXPOPAP were more suitable as they were not correlated to each other. Coefficients of EXPOPAP characterized the relative growth rate per incident PAR integral [$r_{m,i}$ (MJ m⁻²)⁻¹] and light use efficiency (LUE, g MJ⁻¹) at closed canopy. In all four experiments, no interaction effects between treatments on crop growth parameters were found. $r_{m,i}$ and LUE were not different between ASS and natural light treatments, but were increased significantly when light levels were reduced by shading in the summer experiments. There was no consistent effect of plant density on growth parameters. $r_{m,i}$ and LUE showed hyperbolic relationships to average daily incident PAR averaged over 10-d periods after planting ($r_{m,i}$) or before final harvest (LUE). Based on those relationships, maximum relative growth rate (r_m , g g⁻¹ d⁻¹) and maximum crop growth rate (c_m , g m⁻² d⁻¹) were described successfully by rectangular hyperbolic relationships to daily incident PAR. In model validation, total dry mass of shoot (W_{shoot} , g m⁻²) simulated over time was in good agreement with measured ones in three independent experiments, using daily incident PAR and leaf area index as inputs. Based on these results, it is concluded that the expolinear growth equation is a useful tool for quantifying cut chrysanthemum growth parameters and comparing growth parameter values between different treatments, especially when light is the growth-limiting factor. Under controlled environmental conditions the regression model worked satisfactorily, hence the model may be applied as a simple tool for understanding crop growth behaviour under seasonal variation in daily light integral, and for planning cropping systems of year-round cut chrysanthemum. However, further research on leaf area development in cut chrysanthemum is required to advance chrysanthemum crop growth prediction.

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Key words: Assimilation light, chrysanthemum, crop growth model, crop growth rate, expolinear, dry mass production, light use efficiency, plant density, shading, radiation use efficiency, relative growth rate.

INTRODUCTION

Year-round cut chrysanthemum (a short-day plant) is grown in intensive industrialized cultivation systems, with scheduled planting and harvesting throughout the year (Van der Hoeven *et al.*, 1975; Langton *et al.*, 1999; Lee *et al.* 2002). Hence, prediction of growth and development over time is important for maximizing cropping system efficiency in the crop. In complex production systems such as these, crop growth models can be useful tools for optimizing farm management, for integrating knowledge of crop physiology and climatic control, and for controlling crop growth and

decision support systems (Krug, 1989; Challa, 1990; Lentz, 1998; Marcelis *et al.*, 1998; Larsen and Persson, 1999).

Physiological plant processes are too complex to be described by simple models, but simplifications are often useful for interpolating or extrapolating measured data. Theoretically, crop growth rate is approximately linearly related to daily light integral (Challa *et al.*, 1994; Challa and Bakker, 1999), but firm experimental evidence for year-round glasshouse crops is lacking (Challa *et al.*, 1994; Challa and Bakker, 1999). When nutrients, water, pests and diseases are not limiting, potential crop growth rate can be explained mainly by the ability of the crop to intercept and utilize radiation at a closed canopy (De Wit *et al.*, 1978). Crop light use efficiency (LUE, g MJ⁻¹) has been defined as dry mass production per unit of intercepted photosynthetically active radiation (PAR, MJ m⁻²) (Monteith, 1977,

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1994). The LUE concept has often been used for analysing and modelling crop growth (Bonhomme, 2000), due to its simplicity and supporting experimental evidence (Kage *et al.*, 2001). Bonhomme (2000) reported that variation in LUE is considered small if environmental conditions are not seriously growth limiting, such that crop growth rate is directly proportional to light interception. However, Lee *et al.* (2002) found that LUE varied widely between seasons, ranging from 3.4 (summer) to 5.3 (winter) g MJ⁻¹ of intercepted PAR for glasshouse-grown cut chrysanthemum. Moreover, the seasonal variation in LUE raised doubts about the linear relationship between intercepted light integral and accumulated dry mass production in year-round cultivated crops, where growth periods include conditions of seasonally decreasing or increasing daily light integral. Therefore, in year-round cut chrysanthemum, LUE may need to be generalized as a variable rather than a constant value for analysing and modelling crop growth.

Dry mass production of cut chrysanthemums over time as an indicator for crop physical yield has been described effectively by the expolinear growth equation (Goudriaan and Monteith, 1990; Goudriaan, 1994), although with large seasonal variation in crop growth parameters (Lee *et al.*, 2002). The expolinear growth equation has been used in many crops to describe crop growth dynamics and interpret measured data, as the equation contains biologically meaningful parameters, namely maximum relative growth rate (r_m , g g⁻¹ d⁻¹), maximum absolute growth rate (c_m , g m⁻² d⁻¹) and lost time (t_b , d) (Goudriaan and Monteith, 1990; Goudriaan, 1994; Tei *et al.*, 1996; Both *et al.*, 1997; Dennett and Ishag, 1998; Ishag and Dennett, 1998; Monteith, 2000). Lee *et al.* (2002) report that parameter estimation for the expolinear growth equation is problematic when daily light integral varies systematically during the crop growth period, and Goudriaan and Monteith (1990) have worried about using the expolinear model under such conditions. Since variation in crop growth is the final result of variations in environmental conditions throughout the crop growth period, Tei *et al.* (1996) proposed that fitting empirical models to measured dry mass production could improve the accuracy of estimating dependent variables if the independent variable in the model is replaced by environmental time, i.e. considering the interdependent variables daily light integral, temperature integral and CO₂ concentration (Aikman and Scaife, 1993; Tei *et al.*, 1996). Therefore, in the case of modelling crop growth under temporally varying environmental conditions, the problem of estimating parameters for the expolinear growth equation could be solved by considering 'environmental time' instead of days after planting.

The main objective of this study was to develop a regression model with biologically meaningful parameters, and to model the effect of radiation and plant densities on crop growth rate for year-round cut chrysanthemum. For model development, chrysanthemum crops were grown at various plant densities under various light conditions in glasshouse compartments throughout the year. Growth parameters of the expolinear growth equation were estimated as a function of incident PAR integral (EXOPAR) as the only independent variable. In this approach, the

growth parameters r_m , c_m and t_b of the expolinear growth equation can be replaced by relative growth rate per incident PAR integral [$r_{m,i}$ (MJ m⁻²)⁻¹] in the exponential growth phase, light use efficiency (LUE, g MJ⁻¹) in the linear growth phase at full light interception by the crop, and lost incident PAR integral (i_b , MJ m⁻²), respectively. To improve accuracy of estimated growth parameters, the growth equation was reformulated to reduce correlation between parameters. Based on those estimated parameters, an empirical model was developed for daily crop growth rate as a function of daily incident PAR and validated with three independent experiments, of which measured data were not used for model development.

Model description

The expolinear growth model as a function of time (t , d) has three parameters, namely maximum relative growth rate (r_m , g g⁻¹ d⁻¹) in the exponential phase, maximum growth rate (c_m , g m⁻² d⁻¹) in the linear phase, and lost time (t_b , d) to indicate the apparent time lost during canopy development before all radiation is intercepted; it determines the position of the curve on the time-axis (Goudriaan and Monteith, 1990; Goudriaan, 1994). W is the total dry mass of shoots per unit ground area (g m⁻²).

$$W = (c_m/r_m) \times \ln \left[1 + e^{(r_m(t-t_b))} \right] \quad (1)$$

A detailed formulation of expolinear growth (eqn 1) has been presented by Goudriaan and Monteith (1990) and Goudriaan (1994), and the model has been applied to cut chrysanthemum (Lee *et al.*, 2002). Therefore, only the specific approach to estimate the regression coefficients is detailed here. In eqn (1) c_m can be considered, as a first approximation, to be proportional to incident PAR at closed canopy (Goudriaan and Van Laar, 1994). The relationship coefficient between incident PAR (MJ m⁻² d⁻¹) and c_m (g m⁻² d⁻¹), therefore, can be defined as light use efficiency (LUE, g MJ⁻¹) in a closed canopy. Equation (1) was rewritten as a function of the accumulated daily incident PAR integral inside the glasshouse (I_t , integrated PAR, MJ m⁻²), and by substituting the growth parameters r_m , c_m and t_b with their respective analogies.

$$W = (LUE/r_{m,i}) \times \ln \left[1 + e^{(r_{m,i}(I_t-i_b))} \right] \quad (2)$$

where $r_{m,i}$ [(MJ m⁻²)⁻¹] is the relative growth rate per incident PAR integral in the exponential growth phase, LUE (g MJ⁻¹) is light use efficiency in the linear growth phase at closed canopy, and i_b is the lost incident PAR integral (MJ m⁻²). I_t (MJ m⁻²) is the accumulated daily incident PAR integral from planting day to harvest for each periodic harvest. In general, the parameters in eqns (1) and (2) are strongly correlated (Goudriaan and Monteith, 1990; Goudriaan, 1994). In particular, the parameters t_b and i_b strongly correlate with r_m and $r_{m,i}$, respectively

TABLE 1. Basic outlines of seven glasshouse experiments with cut chrysanthemum

Expt	Year	Planting date	No. of long days	No. of short days	Final harvest date	Plant density (plants m ⁻²)	<i>N</i>	Global radiation* (MJ m ⁻² d ⁻¹)	Temp. [†] (min/max) [‡] (°C)	[CO ₂] [‡] (min/max) [§] (μmol mol ⁻¹)	Expt goal
1	2000	12	21	58	91	32, 48, 64	14	2–7	21 (20/22)	415 (374/445)	D
2	1999	273	21	60	354	32, 48, 64	13	6–2	19 (16/20)	432 (345/579)	D
3	1999	126	22	53	201	32, 48, 64	9	17–19	21 (18/28)	407 (357/460)	D
4	2000	160	21	55–60	236–241	32, 64, 80	14	22–15	22 (19/29)	349 (318/413)	D
5	2000	12	21	58–61	91–94	32, 48, 64	14	2–7	17 (17/18)	415 (374/443)	V
6	1999	29	17	56	102	32, 48, 64	8	4–9	19 (17/22)	447 (354/613)	V
7	2000	250–264	7, 14, 21	63	334	64	12	8–2	20 (17/23)	407 (348/471)	V

The experiments were either used for model development (D) or validation (V) of a model for the growth of the crop.

Dates are expressed as day of year (day 1 = 1 January).

N is the number of harvests with 3–12-d intervals during the experimental crop growth period.

* Average values over the first 2 weeks and the last 2 weeks of cultivation.

† Average daily (24 h) means for the whole growing period.

‡ Measured above the crop inside the glasshouse between 1000 and 1600 h, averaged over the whole growing period.

§ Minimum and maximum daily temperatures or [CO₂] during an experiment.

(Goudriaan and Monteith, 1990). Hence, i_b in eqn (2) can be decomposed as

$$i_b = -\ln\left[\frac{f_0}{1-f_0}\right]/r_{m,i} \quad (3)$$

where f_0 is the initial fraction of light interception by the initial leaf area index (LAI₀),

$$f_0 = 1 - e^{kLAI_0} \quad (3a)$$

Therefore, eqn (2) can be rewritten as

$$W = (LUE/r_{m,i}) \times \ln\left[1 + \frac{f_0}{1-f_0} e^{(r_{m,i} \cdot t_i)}\right] \quad (4)$$

At the start, accumulated incident PAR is set to zero assuming no growth at planting date. Hence, initial total dry mass of shoot (W_0) can be derived from eqn (3) by

$$W_0 = (LUE/r_{m,i}) \times \ln\left[1 + \frac{f_0}{1-f_0}\right] \quad (5)$$

The fraction of light (f_0) can be eliminated from eqn (4) because

$$\frac{f_0}{1-f_0} = e^{\frac{W_0 r_{m,i}}{LUE}} - 1 \quad (6)$$

Hence, eqn (2) can be rewritten as

$$W = (LUE/r_{m,i}) \times \ln\left[1 + \left(e^{\frac{W_0 r_{m,i}}{LUE}} - 1\right) \times e^{(r_{m,i} \cdot t_i)}\right] \quad (7)$$

Assuming a constant leaf area ratio (LAR, m² g⁻¹) throughout the crop growth period,

$$r_{m,i}/LUE = kp_i SLA_n = kLAR \quad (8)$$

where k is the light extinction coefficient which is assumed constant at 0.72 (Goudriaan and Van Laar, 1994; Lee *et al.*, 2002), p_i is the fraction of dry matter partitioned into new leaves and SLA_n is the specific leaf area of new leaves (m² g⁻¹). Hence, eqn (7) can be rewritten as

$$W = \left(\frac{1}{kLAR}\right) \ln[1 + (e^{W_0 kLAR} - 1)e^{(C_I kLAR)}] \quad (9)$$

where C_I (g m⁻²) is the accumulated maximum crop growth rate (c_m , g m⁻² d⁻¹) at fully closed canopy. c_m was estimated by multiplying the daily incident PAR (MJ m⁻² d⁻¹) with LUE (see Results). Actual crop growth rate (GR, g m⁻² d⁻¹), therefore, can be calculated as

$$GR = c_m(1 - e^{-kLAI}) \quad (10)$$

Assuming a constant leaf area ratio (LAR) over time, the LAI growth rate can be calculated by multiplying GR with a constant LAR, hence the growth depends purely on GR.

MATERIALS AND METHODS

General experimental set-up

Seven experiments (Table 1) with different planting dates were conducted over 2 years using cut chrysanthemum (*Chrysanthemum morifolium* Ramat.), cultivar 'Reagan Improved' (CBA, Aalsmeer, The Netherlands). Experiments were conducted in two (expts 1, 3, 5 and 7) or three (expts 2, 4 and 6) compartments (12.0 × 12.8 m) of a multispan Venlo-type glasshouse at Wageningen

TABLE 2. Average daily incident of photosynthetically active radiation (PAR) integral in experiments used for the development of a model for cut chrysanthemum growth

Expt	Day interval	Average incident PAR* (MJ m ⁻² d ⁻¹)			Incident PAR from lamps† (μmol m ⁻² s ⁻¹)	
		CON	ASS	SD	INC	HPS
1	0–19	0.49	1.10	0.23	3.5 ± 0.4	48.1 ± 3.2
	20–39	0.93	1.31	0.40		
	40–59	1.30	1.67	0.46		
	60–end	1.47	1.84	0.80		
	Average	1.09	1.48	–		
2	0–19	1.49	1.99	0.47	4.8 ± 0.8	39.6 ± 0.5
	20–39	0.98	1.28	0.39		
	40–59	0.69	0.98	0.26		
	60–end	0.44	0.73	0.21		
	Average	0.89	1.24	–		
3	0–19	3.96	–	1.34	–	–
	20–39	3.92	–	1.20	–	–
	40–59	3.99	–	1.52	–	–
	60–end	4.22	–	1.04	–	–
	Average	4.01	–	–	–	–
4	0–19	4.81	–	1.58	–	–
	20–39	2.96	–	1.11	–	–
	40–59	3.52	–	0.84	–	–
	60–end	3.32	–	0.96	–	–
	Average	3.64	–	–	–	–

* Including amount of light from supplementary lighting: ASS, assimilation lighting by high-pressure sodium lamps (HPS); CON, natural light with (expts 1–2) or without (expts 3–4) supplementary lighting by incandescent lamps (INC); SD, standard deviation.

† Incident PAR from the lamps was measured at about 3.2 m below the lamps inside the glasshouse during night conditions.

University, The Netherlands (lat. 52°N). Experiments 1–4 were used for model development and expts 5–7 for model validation.

Rooted cuttings of cut chrysanthemum were obtained from a commercial propagator (Fides Goldstock Breeding, Massland, The Netherlands) and planted at 32, 48, 64 or 80 plants m⁻² on eight parallel soil beds. A border soil bed was present on both sides of the two experimental beds in each half of the compartments in expts 1, 2, 5 and 6 and at each side of the compartment in expts 3, 4 and 7. General crop management of year-round cut chrysanthemum has been described elsewhere (Lee *et al.*, 2002). Long day (LD) was about 15 h (expts 3 and 4), 16 h (expt 6) or 19 h (expts 1, 2, 5 and 7) for 7–22 d, and short day (SD) was 10 h (expt 6) or 11 h (expts 1–5 and 7) until the end of an experiment. Day length was controlled by incandescent lamps (INC, 4–6 μmol m⁻² s⁻¹), high-pressure sodium lamps (HPS, 42–57 μmol m⁻² s⁻¹, SON-T 400 W, Philips, Amsterdam, The Netherlands) and blackout screens. Lamps were on continuously during day h of LD and SD periods (expts 1, 2, 5 and 6) or were controlled based on outside global radiation (expt 7). In the latter case, lamps were turned on when outside global radiation was less than 150 W m⁻² and turned off when it was more than 250 W m⁻². In expts 3 and 4, the same white screens with two different transmissivities were used for shading the crops on two soil beds each. Shading screens were placed 1.5 m above the soil surface on fixed rectangular frames. An irrigation pipe was placed on each soil bed (expt 3), or both on soil beds and under the shading screens (expt 4), in each compartment.

INC and HPS lamps (3.2 m above soil surface) were installed in half a compartment, except in expt 7 where only HPS lamps were installed in a compartment. Lighting contamination between compartments was minimized by installing the same INC or HPS lamps in adjacent compartments, and within compartments aluminium foil (30 cm length) was attached to half the luminaries of HPS lamps near INC lighting plots. Adjusting the position and distance of reflectors to ensure homogeneous light distribution to the crops, light intensity was measured during late evening by a 1 m line quantum sensor (LI-191 SA; LI-COR, Lincoln, NB, USA) in 18 equally spaced places at crop level after planting (Table 2). In expt 3, light intensity above and below the shading screens was measured in 27 places (equally spaced) using a 1 m line quantum sensor (LI-191 SA). In expt 4, two quantum sensors (LI-190 SA; LI-COR, Lincoln, NB, USA) were placed permanently above and below each type of screen and read every 5 min in one of the compartments. The shading ratio in expts 3 and 4 was taken on a coefficient of linear regression between light intensities above and below the screen. Light transmission of the two shading screens was 65 % ± 3.4 % and 45 % ± 3.0 % for expt 3 and 66 % ± 1.6 % and 42 % ± 1.6 % for expt 4. Since the difference between the transmissions in two experiments was marginal, the values 66 % ± 1.9 % and 43 % ± 1.9 % averaged over the two experiments were used. Daily incident PAR inside the glasshouse compartment (MJ m⁻² d⁻¹) was calculated according to Bot (1983), Gijzen (1992), Heuvelink *et al.* (1995) and Lee *et al.* (2002). More detailed information about the calculation of incident PAR

TABLE 3. Pearson's correlation coefficients (sample size = 90) for estimated parameters of eqns (1), (2), (4) and (7)

Eqn	Parameters	$r_{m,i}$	LUE	r_m	c_m
1	c_m	—	—	-0.222*	1.000
	t_b	—	—	-0.770**	0.313**
2	LUE	0.043	1.000	—	—
	t_b	-0.811**	-0.235*	—	—
4	LUE	0.043	1.000	—	—
	F_0	0.711**	-0.023	—	—
7	LUE	0.065	1.000	—	—
	W_0	-0.139	0.176	—	—

** Correlation is significant at the 0.01 level (two-tailed).

* Correlation is significant at the 0.05 level (two-tailed).

in these glasshouse compartments has been described elsewhere (Lee *et al.*, 2002). Additional light from the lamps and light reduction by SD treatments were included in the daily incident PAR integral. Daily incident PAR under shading screens is a proportion of daily light integral of non-shading treatment (100%) in expts 3 and 4.

Day/night temperature set point was 21/20 °C for expt 1, 17/16 °C for expt 5 and 18/19 °C for the other experiments. The ventilation temperature was 1 °C higher than the indicated day/night temperature set point. CO₂ concentration in each compartment was measured by a CO₂ analyser (URAS G; Hartman & Braun, Frankfurt, Germany) and was maintained at a value of at least 350 µmol mol⁻¹ by pure CO₂ enrichment in all experiments except expt 2, where CO₂ concentration was maintained at 400 µmol mol⁻¹ in two compartments. Daily outside global radiation (Solarimeter; Kipp & Zonen, Delft, The Netherlands) was obtained from a meteorological station located at about 100 m distance from the glasshouse compartments. Glasshouse temperature (PT 500 element) and CO₂ concentration were recorded every 5 min by a commercial VitaCo climatic control system (Hoogendoorn, Vlaardingen, The Netherlands) (Table 1).

Experimental treatments and experimental design

Experiments 1, 2 and 5 were conducted in two and expt 6 in three compartments with two levels of controlled lighting (CON provided by INC) and supplementary assimilation lighting (ASS provided by HPS) in each half of a compartment. Within each lighting plot, three plant densities (32, 48 and 64 plants m⁻²) were distributed randomly on two soil beds in each half of a compartment according to a split plot design. Experiments 3 and 4 were conducted in three compartments with three light levels, namely 100 % natural light, 66 % and 34% of natural light in each compartment, and within each light level three plant densities (32, 48 and 64 plants m⁻² for expt 3 and 32, 64 and 80 plants m⁻² for expt 4) were distributed randomly on two soil beds according to a split-plot design. Experiment 7 was conducted in two compartments under ASS lighting (HPS; 56.6 ± 0.8 µmol m⁻² s⁻¹) with three different durations of LD (experimental unit) obtained by three

planting dates at 1-week intervals as a complete randomized design. Plants were planted at 64 m⁻².

Plant measurements

Destructive samplings were carried out every 3–12 d until final harvest. Samples were taken from five or six plants per experimental plot amongst those surrounded by border plants in two rows on each side of a bed. Total leaf area per plant was measured (LI-3100 area meter; LI-COR, Lincoln, NB, USA), as well as fresh and dry mass (dried at 105 °C for 14 h in a ventilated oven) of leaves (including petioles), stems and flowers (including buds). No root measurements were made. All measurements were done on individual plants for each experimental plot in each compartment (replication).

Model validation

Three independent experiments were used for crop growth model validation using measured initial total dry mass of shoot (W_0 , g m⁻²), leaf area index (LAI) and daily incident PAR (I , MJ m⁻² d⁻¹). As mentioned in the model description, leaf area ratio (LAR) is defined by specific leaf area (SLA_n) and fraction of dry matter partitioned into leaves (p_1). p_1 varies strongly throughout plant development (Hughes and Cockshull, 1972; Karlsson and Heins, 1992) whereas SLA_n varies with temperature and incident PAR (Acock *et al.*, 1979). Therefore, it may not be possible to ignore variation in LAR to calculate LAI dynamics for flowering cut chrysanthemum. To address this problem, total dry-mass production in time was calculated as accumulated GR (eqn 10) with initial total dry mass of shoot as input. Microsoft Excel 2000 (Microsoft Corporation, Washington, USA) was used for model validation using the Euler's integration method. In addition, Prism (GraphPad Software Inc., San Diego, USA) was used to fit measured LAI by a cubic spline curve (Motulsky, 1999), in order to obtain the daily fraction of light intercepted by the crop for eqn (10).

Statistical analyses of measurements and model parameters

Regression coefficients for eqns (1), (2), (4) and (7) were estimated by minimizing the residual sum of squares (least squares method), using the non-linear fitting procedure of the SPSS package (SPSS Inc, Chicago, USA). Using SPSS, Pearson's correlation coefficients were estimated to select the best equation for estimating crop growth parameters, i.e. the equation with no significant correlations between estimated parameters. Of the selected equation, estimated parameters $r_{m,i}$ and LUE were subjected to treatment effect tests. Effects of light regime, plant density and their interaction were tested by ANOVA based on the experimental design for expts 1–4 using the statistical software package Genstat 5 (VSN International Ltd, Herts, UK). Both light regime and plant density effects were separated in a linear and a quadratic component. For significant ($P < 0.05$) effects, means were separated by Student's *t*-tests ($P = 0.05$) using least significant differences (LSD).

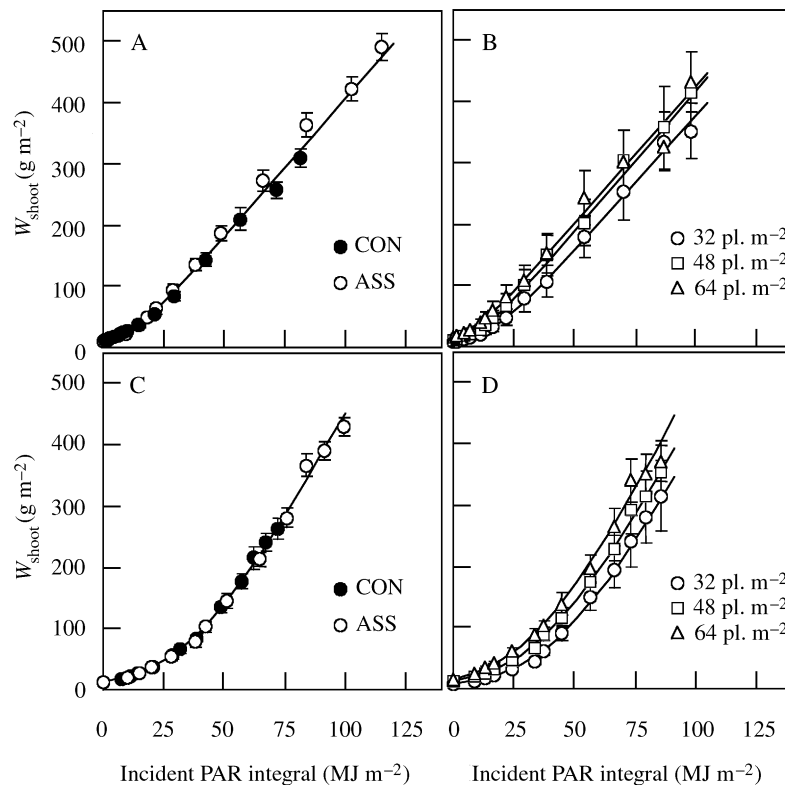


FIG. 1. Dynamics of total shoot dry mass (W_{shoot}) of cut chrysanthemum as a function of incident PAR integral under control (CON) and assimilate lighting (ASS, provided by high-pressure sodium lamps) (A) and at different plant densities (B) for crops planted in January (A and B: expt 1) and September (C and D: expt 2). Vertical bars indicate standard errors of mean measured W_{shoot} .

As mentioned before, $r_{m,i}$ defines the exponential growth phase and LUE the linear growth phase at closed canopy. Therefore, estimated $r_{m,i}$ and LUE were plotted against daily incident PAR averaged over 10 d from planting dates ($r_{m,i}$) or before final harvest (LUE) instead of over the whole growing period. Although the number of days for averaging incident PAR was arbitrary, the hyperbolic relationship to incident PAR gave a good description between averaged incident PAR and $r_{m,i}$ or c_m .

RESULTS

Fitting of expolinear growth equations

Fitting of the expolinear equation as a function of time (eqn 1) and as a function of incident PAR integral (eqn 2) described effectively the observed total shoot dry mass (W_{shoot}) in all experiments. Although the fitting of eqn (1) to periodically measured W_{shoot} accounted for over 98 % of the variance, its estimated parameters were correlated strongly with each other (Table 3). Furthermore, these estimates were particularly biased when daily light integral changed drastically during the crop growth period in expts 1 and 2. These biases can be eliminated when fitting the model as a function of incident PAR integral (eqns 2, 4 and 7). Although the fitting of either eqns (2) or (4) adequately described W_{shoot} dynamics as a function of incident PAR

integral, strong correlations remained between $r_{m,i}$ and i_b (eqn 2) and $r_{m,i}$ and f_0 (eqn 4) (Table 3). Therefore, measured W_{shoot} was fitted as a function of daily incident PAR by eqn (7) (EXPOPARG), to analyse and generalize estimated crop growth parameters without significant correlation between parameters (Table 3).

Estimated growth parameters

Total shoot dry mass (W_{shoot}) dynamics as a function of incident PAR integral followed a pattern of expolinear growth in all experiments (Figs 1 and 2). Estimated initial W_{shoot} was identical to measured initial W_{shoot} in all experiments. Since there were no significant correlations between parameters of eqn (7) (Table 3), treatment effects on estimated parameter values could be determined unambiguously (Tables 4 and 5). In all four experiments, no interaction effects between treatments were observed. In expts 1 and 2, there were no significant effects of assimilation lighting and plant density on crop growth parameters (Table 4). However, the overall average of $r_{m,i}$ appeared to be higher in expt 1 than in expt 2, whereas LUE was higher in expt 2 than in expt 1 (Table 4), indicating a seasonal effect on crop growth parameters. In expts 3 and 4, $r_{m,i}$ and LUE were significantly different between light levels, and increased linearly with decreasing light levels

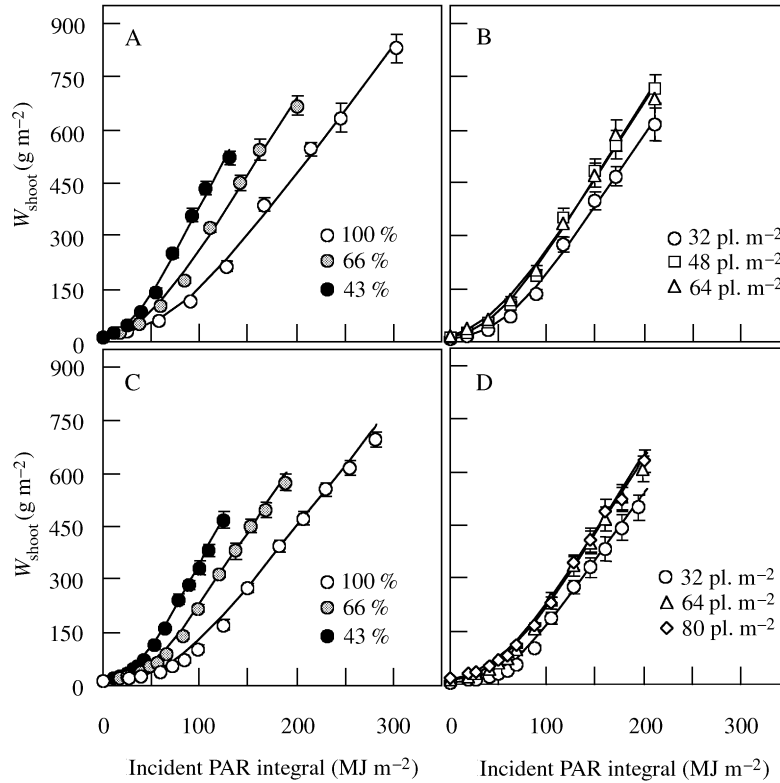


FIG. 2. Dynamics of total shoot dry mass (W_{shoot}) of cut chrysanthemum as a function of incident PAR integral under different shading treatments (A and C) at different plant densities (B and D) for crops planted in May (A and B: expt 3) and June (C and D: expt 4). Vertical bars indicate standard errors of mean measured W_{shoot} .

(Table 5). No plant density effects on growth parameters were detected in expt 3, whereas $r_{m,i}$ and LUE at 32 plants m⁻² were significantly different from $r_{m,i}$ and LUE at the highest plant densities in expt 4 (Table 5). In the latter experiment, crop growth parameters appeared as a linear trend to plant density (Table 5).

Generalization of $r_{m,i}$ and LUE parameters

$r_{m,i}$ and LUE showed hyperbolic relationships to averaged daily incident PAR (Fig. 3), which accounted for 52 and 66 % of variance in $r_{m,i}$ and LUE, respectively. Furthermore the ratio between the regressions of $r_{m,i}$ and LUE in Fig. 3 is identical to the product of k LAR_0 as mentioned before (Goudriaan and Monteith, 1990). Therefore, the initial LAR of 0.0327 m² g⁻¹ can be derived from the ratio between the two non-linear regressions when daily incident PAR equals zero. As daily light integral varies largely with season in northern Europe, seasonal variation in LAR_0 for year-round cut chrysanthemum might be described by a rectangular hyperbola as function of daily incident PAR (I , MJ m⁻² d⁻¹).

$$LAR_0 = a + \frac{b}{(1 + dI)} \quad (11)$$

where $a = 0.0109$ m² g⁻¹, $b = 0.0218$ m² g⁻¹ and $d = 0.9186$ (MJ m⁻² d⁻¹)⁻¹. Hence seasonal variation in LAR_0 can be

described as a function of daily incident PAR throughout the year.

The maximum relative growth rate (r_m , d⁻¹) at early growth phase and maximum crop growth rate (c_m , g m⁻² d⁻¹) at fully intercepted incident PAR can be extrapolated from the regressions of $r_{m,i}$ and LUE (Fig. 3) by multiplying with daily incident PAR (I , MJ m⁻² d⁻¹). In Fig. 3, regressions of $r_{m,i}$ and LUE are, therefore, identical to the ratios r_m/I and c_m/I , respectively. Maximum relative growth rate (r_m) as a function of daily incident PAR (I , MJ m⁻² d⁻¹) can be formulated as

$$r_m = \frac{r_{m,i0} R_i I}{(R_i + r_{m,i0} I)} \quad (12)$$

where $r_{m,i0}$ is initial relative growth rate per incident PAR integral, equal to 0.1813 g g⁻¹ [MJ m⁻²]⁻¹, and R_i is the intrinsic maximum relative growth rate (d⁻¹), equal to 0.1973 g g⁻¹ d⁻¹ (Fig. 4).

Using the same approach as with r_m , maximum crop growth rate (c_m) as a function of daily incident PAR (I , MJ m⁻² d⁻¹) can be formulated as

$$c_m = \frac{LUE_0 C_i I}{(C_i + LUE_0 I)} \quad (13)$$

where LUE_0 is initial LUE, equal to 7.70 g MJ⁻¹, and C_i is the intrinsic maximum crop growth rate, equal to 25.18 g

TABLE 4. Effects of supplementary lighting and plant density on relative growth rate per incident PAR integral ($r_{m,i}$) in the exponential growth phase, and on light use efficiency (LUE) in the linear growth phase at closed canopy, estimated by the expolinear growth equation as a function of accumulated incident PAR integral for chrysanthemum crops planted in January 2000 (expt 1) and September 1999 (expt 2)

Treatment	Expt 1		Expt 2	
	LUE (g MJ ⁻¹)	$r_{m,i}$ (MJ m ⁻²) ⁻¹	LUE (g MJ ⁻¹)	$r_{m,i}$ (MJ m ⁻²) ⁻¹
Light*				
CON	4.33	0.1176	6.66	0.0620
ASS	4.72	0.1169	6.72	0.0629
Plant density				
32	4.44	0.1104	6.50	0.0626
48	4.62	0.1165	6.67	0.0622
64	4.51	0.1249	6.91	0.0626
LSD†	0.55	0.0532	0.72	0.0079
F-probabilities				
Light	0.149	0.497	0.926	0.857
Density	0.703	0.691	0.368	0.877
Linear	0.734	0.421	0.184	0.830
Quadratic	0.470	0.936	0.872	0.664
Light × density	0.098	0.623	0.063	0.056

* CON, control lighting by incandescent lamps; ASS, assimilation lighting by high-pressure sodium lamps.

† Least significant difference ($P = 0.05$) between means.

TABLE 5. Effects of light level and plant density on relative growth rate per incident PAR integral ($r_{m,i}$) in the exponential growth phase, and on light use efficiency (LUE) in the linear growth phase at closed canopy, estimated by the expolinear growth equation as a function of accumulated incident PAR integral for chrysanthemum crops planted in May 1999 (expt 3) and June 2000 (expt 4)

Treatment	Expt 3		Expt 4	
	LUE (g MJ ⁻¹)	$r_{m,i}$ (MJ m ⁻²) ⁻¹	LUE (g MJ ⁻¹)	$r_{m,i}$ (MJ m ⁻²) ⁻¹
Light				
100%	3.63	0.0344	3.47	0.0401
66%*	4.34	0.0497	4.23	0.0520
43%*	5.23	0.0694	5.44	0.0639
LSD	0.50	0.0064	0.54	0.0093
Plant density				
32	4.22	0.0517	3.85	0.0652
48	4.68	0.0511	—	—
64	4.30	0.0507	4.55	0.0479
80	—	—	4.74	0.0429
LSD†	0.55	0.0072	0.52	0.0075
F-probabilities				
Light	<0.001	<0.001	<0.001	0.002
Linear	<0.001	<0.001	<0.001	<0.001
Quadratic	0.212	0.053	0.076	0.509
Density	0.196	0.959	0.008	<0.001
Linear	0.751	0.781	0.002	<0.001
Quadratic	0.081	0.961	0.639	0.442
Light × density	0.055	0.130	0.155	0.742

* Proportion of incident PAR transmitted through two shading screens.

† Least significant difference ($P = 0.05$) between means.

m⁻² d⁻¹, which can be limited by factors other than radiation, namely CO₂ concentration at high daily light integral and temperature (Fig. 4).

Based on the theoretical relationship between $r_{m,i}$ and LUE, eqn (9) may be derived from eqn (7). Assuming a constant LAR, growth rate of LAI is proportional to crop growth rate. Hence, total dry mass of shoot (W_{shoot} , g m⁻²) calculated from eqn (9) provides an accurate value, as

accumulated crop growth rate in eqn (10) is based on simulated LAI, at constant LAR over growth period (eqn 11) (data not shown).

Model validation

In flowering chrysanthemum, fraction p_1 decreased systematically with increased total dry mass (Hughes and

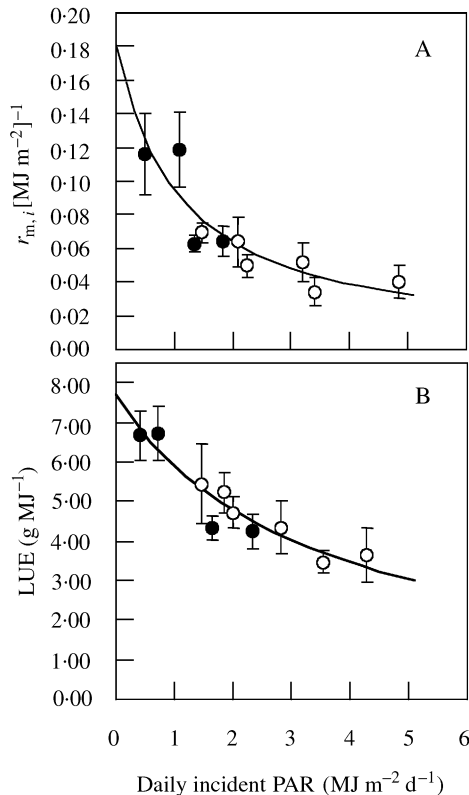


FIG. 3. Relative growth rate per incident PAR integral ($r_{m,i}$, A) and light use efficiency (LUE, B) for a cut chrysanthemum crop at different levels of daily incident PAR. Vertical bars indicate standard deviation of parameter estimates ($n = 6$ for closed symbols, $n = 9$ for open symbols). Daily incident PAR (I) in the period after planting ($r_{m,i}$) or before final harvest (LUE) is averaged over 10 d periods. Regressions are: A, $1/r_{m,i} = 5.5172 (\pm 1.344) + 5.0679 (\pm 0.505) \times I$, $R^2 = 0.52$, $n = 78$; and B, $1/\text{LUE} = 0.1299 (\pm 0.008) + 0.0397 (\pm 0.003) \times I$, $R^2 = 0.66$, $n = 78$.

Cockshull, 1972), whereas in vegetatively grown chrysanthemum SLA varied with temperature and incident PAR under closed canopy (Acock *et al.*, 1979). Therefore, p_l and SLA were not suitable as constants for model validation in this study. Alternatively, measured LAI was used as input for validating the generalized model for crop growth rate formulated by eqn (10). As inputting measured daily incident PAR and leaf area index in expts 5–7 (Table 1), simulated crop growth patterns adequately described measured W_{shoot} in all independent experiments (Fig. 5). In expt 5, estimated W_{shoot} was higher than measured ones for the crop grown at 17 °C under natural light conditions, whereas they were almost identical for the crop grown at the same temperature under assimilation lighting.

DISCUSSION

The expolinear growth equation as a function of incident PAR integral (EXPOPARG) described periodically measured total shoot dry mass (W_{shoot} , g m⁻²) significantly better than the equation as a function of time (EXPOT), for all experiments, and particularly for expt 2 (data not shown). A similar approach to fitting empirical models has been used

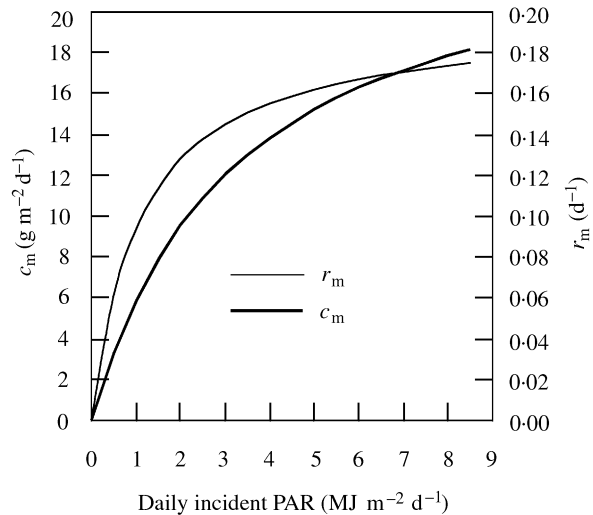


FIG. 4. Maximum relative growth rate (r_m , d⁻¹) and maximum crop growth rate (c_m , g m⁻² d⁻¹) for a cut chrysanthemum crop at different levels of daily incident PAR. Rectangular hyperbolic relationship to daily incident PAR is derived from the hyperbolic relationship of $r_{m,i}$ and LUE to daily incident PAR in Fig. 3. Regressions are, $r_m = 0.1813 \times 0.1973 \times I / (0.1973 + 0.1813 \times I)$, and $c_m = 7.7 \times 25.18 \times I / (25.18 + 7.7 \times I)$.

for describing dry mass dynamics of lettuce, onion and red beet (Tei *et al.*, 1996).

Using EXPOPARG (eqn 7), total shoot dry mass dynamics as related to accumulated daily incident PAR were described effectively. Crop growth parameters were estimated successfully without significant correlations between parameters. Treatment effects on estimated parameters, therefore, could be tested unambiguously in this study (Tables 4 and 5).

Relative growth rate per incident PAR integral in the exponential growth phase ($r_{m,i}$) and light use efficiency in the linear growth phase of the crop (LUE) were explained effectively by hyperbolic relationships to averaged daily incident PAR (Fig. 3). It is worthwhile to compare LUE as estimated by EXPOPARG with LUE as estimated by the slope of the linear relationship between accumulated light integral and accumulated dry mass production (LUE_{IN}). In the summer experiment conducted by Lee *et al.* (2002), LUE estimated by EXPOPARG (3.6 g MJ⁻¹) was almost consistent with LUE_{IN} (3.4 g MJ⁻¹). However, in the control treatment of expt 2 in the present study, LUE estimated by EXPOPARG (6.7 g MJ⁻¹) (Table 5) was higher than LUE_{IN} (5.3 g MJ⁻¹), a discrepancy similar to results of Lee *et al.* (2002) for an experiment in which maximum LAI averaged over three plant densities was 4–8. Based on the overall relationship between LUE and daily incident PAR in this study, and evidence of variation in LUE in a range of incident PAR of 5–10 MJ m⁻² d⁻¹ for cauliflower (Kage *et al.*, 2001), linear regression may underestimate LUE_{IN} at closed canopy, when daily light integral decreases greatly, such as in expt 2 (Table 2). When daily incident PAR increases greatly, such as in expt 1, LUE_{IN} may be overestimated. Therefore, LUE can have non-linear relationships with intercepted PAR

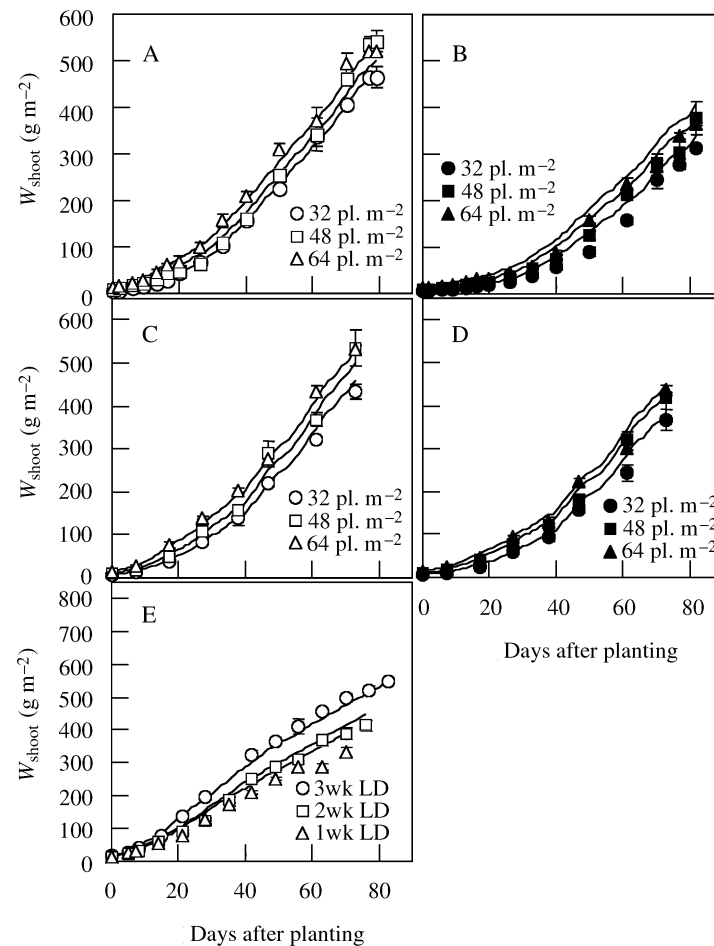


FIG. 5. Measured and simulated dynamics of total shoot dry mass (W_{shoot}) of cut chrysanthemum crops grown under various light conditions (open symbols, ASS supplemented by high-pressure sodium lamps; closed symbols, control lighting supplemented by incandescent lamps), at three plant densities in expts 5 (A and B) and 6 (C and D) and three long-day (LD) periods in expt 7 (E). Actual crop growth rate was calculated using eqn (10) as daily incident PAR and measured leaf area index from the results of three independent experiments. Vertical bars indicate standard errors of mean measured W_{shoot} .

integral and accumulated dry mass production when daily light integral varies systematically during the crop growth period (Table 2).

Due to the hyperbolic relationships of $r_{m,i}$ and LUE to daily incident PAR, maximum relative growth rate (r_m) and maximum crop growth rate (c_m) have a rectangular hyperbolic relation to daily incident PAR integral. A similar response of relative growth rate (RGR) to incident PAR integral was found in young tomato plants, cucumber and sweet pepper (Nilwik, 1981; Bruggink and Heuvelink, 1987; Bruggink, 1992; Challa *et al.*, 1994). From those studies, it was concluded that variation in RGR was due mainly to variation in LAR. Moreover, based on results of the present study and work by other authors (Challa and Schapendonk, 1984; Karlsson *et al.*, 1987; Cockshull *et al.*, 1992; Olesen and Grevsen, 1997; Kage *et al.*, 2001), it can be concluded that crop growth rate is dependent on daily incident PAR and not proportional to light. The saturation response to incident PAR of potential crop growth rate at closed canopy is to be expected, as light saturation of

photosynthesis at leaf level is reflected at crop level. Furthermore, at increased light level in summer the fraction of direct light is greater, but used less efficiently compared with the different radiation of the winter season (Gijzen, 1992). Moreover, at low daily PAR in low light regions, the role of maintenance respiration (R_m) relative to crop gross assimilation rate ($P_{g,c,d}$) should be considered (Heuvelink *et al.*, 2002). R_m is not affected by assimilation light (Heuvelink *et al.*, 2002). If R_m uses a large part ($\geq 50\%$) of $P_{g,c,d}$, a proportional increase in $P_{g,c,d}$ will result in a more than proportional increase in growth, which is proportional to $P_{g,c,d}$ minus R_m (Heuvelink *et al.*, 2002). Cockshull *et al.* (1992) found for tomato crops that loss of dry mass was less than relative loss of light by shading. In the present study, a similar response to reduced light level by shading screens was found in the summer experiments (Fig. 2). This response could be due to improved LUE when shading reduces daily light integral.

Based on the rectangular hyperbolic relation to daily incident PAR, crop growth rate can be used to simulate dry

mass production of year-round cut chrysanthemum crops. Although many authors have found effects of temperature and CO₂ concentration on crop growth, the model developed in the present study is a function of light only. In addition, this model assumes that the light compensation point is zero. According to Penning de Vries and Van Laar (1982), it may be assumed that maintenance respiration equals 1.5 % of dry mass per day. When LAR and LUE are considered as the values of LAR_0 and LUE_0 in the present study, the light compensation point calculated as 0.08 MJ m⁻² d⁻¹ (Bruggink and Heuvelink, 1987). As this value is negligible compared with daily incident PAR integral in the present study, the model assumption of a light compensation point of zero appears acceptable. Model validation using independent experiments effectively simulated dry mass production using measured LAI, initial dry mass and daily incident PAR as input data. The only overestimation of dry mass production occurred when the model was applied to the crop grown at lower temperature and lower daily incident PAR (expt 5). This discrepancy may be due to erratic generalization of LUE through extrapolation of the hyperbolic relation in the lower light range, particularly as LUE varied largely in this range (expt 2). In addition, it may be due to possible variation in the light extinction coefficient between seasons or during the crop growth period.

Equation (9) may be used as a simple method for calculating dry mass production under the assumption that LAR is constant throughout crop growth. Under the same assumption, the growth rate of LAI may be simulated by the variation in crop growth rate. In fact, results of simulating total shoot dry mass (W_{shoot} , g m⁻²), assuming a constant LAR, were not different from simulating W_{shoot} with input of LAI (data not shown). LAR is strongly correlated with light as well as with crop growth stage, due to the variation in SLA and dry matter partitioning into leaves. Dry mass partitioning into leaves has been found to vary greatly with plant dry mass in flowering chrysanthemum (Hughes and Cockshull, 1972), whereas it was shown to remain almost constant under large variation in temperature and daily light integral in vegetatively grown chrysanthemum (Acock *et al.*, 1979). Furthermore, SLA may be strongly correlated with light and temperature (Acock *et al.*, 1979). Prediction of LAI is a crucial subject in crop growth modelling, as the accurate estimation of crop growth rate depends on it. Further research on leaf area development in cut chrysanthemum is required to enable development of an LAI model for this crop.

Given these considerations, the expolinear growth equation proved to be a useful tool for analysing measured total shoot dry mass of cut chrysanthemum and for simulating total shoot dry mass in time. In the present work, cut chrysanthemums were harvested before senescence and measured in terms of total plant dry mass. Hence growth components of relative growth rate and crop growth rate could easily be assessed by fitting the equation to periodically measured total shoot dry mass under constant light conditions during crop growth. However, if systematic variation in environmental factors occurs, growth parameters may be estimated using environmental time as an independent variable.

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ABBREVIATIONS

W, W_{shoot}	Total shoot dry mass	g m ⁻²
r_m	Maximum relative growth rate	d ⁻¹
c_m	Maximum absolute growth rate	g m ⁻² d ⁻¹
t_b	Lost time to attain canopy closure	d ⁻¹
i_b	Lost amount of incident PAR	MJ ⁻¹
pl	Fraction of dry mass partitioning to leaves	—
$r_{m,i0}$	Initial maximum relative growth rate per incident PAR integral	(MJ m ⁻²) ⁻¹
$r_{m,i}$	Maximum relative growth rate per incident PAR integral	(MJ m ⁻²) ⁻¹
R_i	Intrinsic relative growth rate	d ⁻¹
C_i	Integrated maximum crop growth rate	g m ⁻²
C_i	Intrinsic crop growth rate	g m ⁻² d ⁻¹
LUE_0	Initial light use efficiency	g MJ ⁻¹
LUE	Light use efficiency estimated by expolinear equation	g MJ ⁻¹
LUE_{IN}	Coefficient of linear relationship between intercepted PAR integral and total dry mass of shoot	g MJ ⁻¹
PAR	Photosynthetically active radiation	MJ m ⁻² d ⁻¹
I	Daily incident PAR integral	MJ m ⁻² d ⁻¹
I_I	Integrated daily PAR integral	MJ m ⁻²
LAR_0	Initial leaf area ratio	m ² g ⁻¹
LAR	Leaf area ratio	m ² g ⁻¹
LAI	Leaf area index	—
SLA_n	Specific leaf area of new leaves	m ² g ⁻¹
EXPOT	Expolinear growth equation as a function of time	g m ⁻²
EXPOPAR	Expolinear growth equation as function of I	g m ⁻²
R_m	Maintenance respiration rate	g m ⁻² d ⁻¹
$P_{gc,d}$	Gross assimilation rate	g m ⁻² d ⁻¹

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